Lab 2 - A Descriptive Analysis of Travel and Competitive Performance in the NBA

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Introduction

Professional sports leagues such as the NBA require teams to travel frequently across large geographic distances, often across multiple time zones, within tightly scheduled seasons. Travel has long been discussed as a potential contributor to competitive imbalance, fatigue, and home-court advantage, yet the extent to which travel-related factors are reflected in game outcomes remains unclear. Understanding these patterns can help coaches optimize scheduling and travel logistics, assist players and athletic trainers in preparing for physical demands, and inform sports bettors' predictions.

We propose to examine whether the distance an away team travels affects point differential (away team score minus home team score). Our work builds upon previous research, including Deddens and Steenland (1997), which found limited effects of travel distance in their analysis of regular season NBA games. We aim to revisit this question by leveraging a comprehensive dataset spanning across seven decades to provide a more complete picture of travel-related performance patterns.

Data

Data Source - The data set was sourced from Kaggle's "NBA Dataset - Box Scores & Stats". This data set contains data on NBA games from 1947 to 2025 and is updated regularly with official statistics from the NBA's website. It contains 72,278 games and 17 variables such as game attendance and points for each team.

Data Wrangling - The data set was restricted to regular season games, reducing the size by 10%. 71 games were removed where the city was indeterminate, such as the historic mid-west Tri-Cities team. Games with travel distances larger than 3500 miles or which had attendance above 100,000 were filtered from the dataset to focus on US travel and remove anomalies. For each unit of observation, seven additional variables were calculated: (1) the point differential between the away and home teams, (2) home team latitude, (3) home team longitude, (4) away team latitude, (5) away team longitude, (6) the geodesic distance between the home and away team cities, and finally (7) a categorical variable indicating coast to coast travel for the away team.

This categorical variable contains 3 possibilities: travel from east to west coast, travel from west to east coast, or neither, which encompasses all travel within the same region or travel between the central regions of the US and coasts. The point differential was calculated by subtracting the home team's score from the away team's score, so that a positive point differential translates to a win for the away team. Lastly, the latitude and longitude of cities were calculated using geocodes from OpenStreetMap based on the cleaned city name for each team.

After wrangling, the final data set consisted of 67,004 games, of which 30% (20,101) were randomly selected for exploratory data analysis and 70% (46,903) were set aside for modeling.

Operationalization - To capture the relationship between away team travel and game outcomes, the following variables were included in our analysis:

- Point Differential (Output): Captures the direction and magnitude of the game outcome.
- Distance: The geodesic distance traveled by the away team.
- Travel Direction: Tracks the direction of movement and if it was from coast-to-coast.
- Attendance: Game attendance, included as a proxy for the intensity of the home-court environment faced by the away team.
- Year: The year in which the game was played, included to account for potential temporal effects on point differential (see Appendix Figure 2).

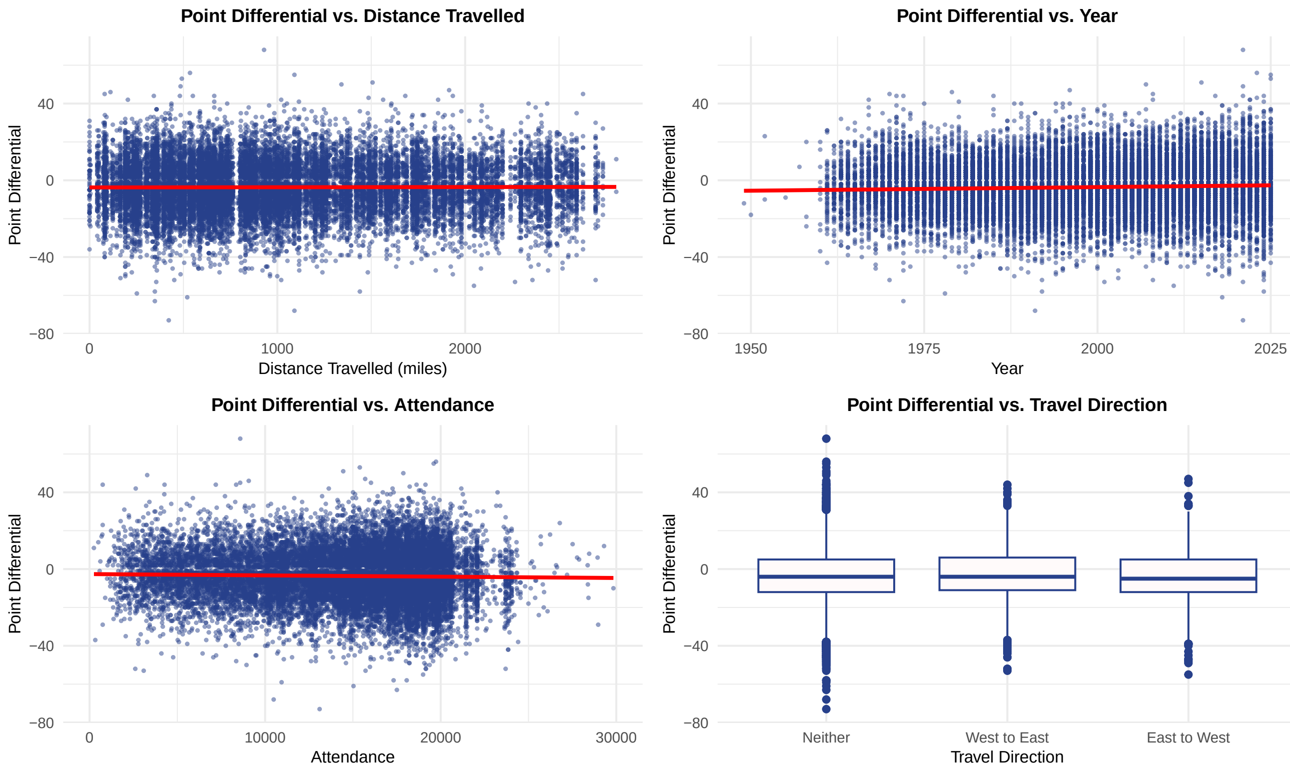

Figure 1: Exploratory data analysis of point differential and independent variables.

Exploratory Data Analysis - Before modeling, the relationships between the input and response variables were explored. From initial exploration, the scatter plot of point differential vs. travel distance shows no clear trend, with much noise around zero. This suggests a weak descriptive relationship between the two variables, and it is likely that other factors play a much larger role in determining game outcomes.

The scatter plot of point differential and year show that the range of the point differentials has gradually increased over time, which can be seen from the fan shape. This suggests that the distribution of game outcomes has widened in more recent years, potentially reflecting structural changes in the NBA which have led to more pronounced wins and losses.

For point differential and attendance, the spread of point differentials increases with attendance up to approximately 20,000 spectators. This pattern suggests greater variability in game outcomes among games with higher attendance.

Finally, the box plots comparing point differential across the three travel direction categories show that west to east travel is associated with a slightly higher median point differential for away teams compared to other categories. This suggests that travel direction may be meaningfully related to away-team performance, motivating its inclusion in the analysis.

Modeling

Model Specification

Given the large sample size, we use a linear regression model with robust standard errors to examine the relationship between point differential and variables such as travel distance, travel direction, game attendance, and year. The regression table below shows four models that progressively add more descriptors.

Model 1: The first model includes only travel distance. The coefficient for this variable is not statistically significant, suggesting that travel distance has little effect on point differential.

Model 2: In this model, we add travel direction (with "Neither" as the reference variable). The coefficients for West-to-East and East-to-West are statistically significant, indicating that teams traveling from the West Coast to the East Coast tend to perform better compared to those traveling from the East Coast to the West Coast.

Model 3: Game attendance is introduced. Although it is statistically significant, the effect is small, suggesting that larger crowds slightly affect the point differential but with limited practical significance..

Model 4: Finally, we add year. We observe a statistically significant positive effect, implying that point differentials have increased over time.

Table 1: Linear Regression Co-efficients for Point Differential Against Input Variables

	Dependent variable:			
	Point Differential			
	(1)	(2)	(3)	(4)
Distance Travelled (miles)	−0.0001	−0.0001	−0.0001	−0.0002
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
West to East Direction		0.640^{**}	0.573^{**}	0.959^{***}
		(0.278)	(0.279)	(0.279)
East to West Direction		−0.448	-0.509^{*}	−0.099
		(0.278)	(0.278)	(0.279)
Game Attendance			-0.00004^{***}	-0.0002^{***}
			(0.00001)	(0.00002)
Year				0.069^{***}
				(0.005)
Constant	-3.380^{***}	-3.355^{***}	-2.814^{***}	-137.748^{***}
	(0.113)	(0.131)	(0.221)	(9.201)
Observations	46,903	46,903	46,903	46,903
R^2	0.00001	0.0004	0.001	0.005
Adjusted R^2	−0.00001	0.0003	0.0005	0.005
Residual Std. Error	13.261 (df = 46901)	13.259 (df = 46899)	13.258 (df = 46898)	13.225 (df = 46897)
Note:				*p<0.1; **p<0.05; ***p<0.01

Model Assumptions

Since the model is a large sample linear model, the required assumptions are as follows:

Independent & Identically Distributed Errors - The linear regression framework assumes that error terms are independent and identically distributed. In this context, games are not fully independent, as team performance may be correlated across games due to persistent team quality, opponent strength, injuries, and broader league dynamics. While including year as a numeric covariate captures long-run temporal trends in point differentials, it does not fully eliminate within-team or within-season dependence. As a result, the model should be interpreted descriptively, with coefficient estimates reflecting average associations rather than causal effects.

A Unique Best Linear Predictor Exists - Covariance and correlation matrices (Appendix Table 2 and 3) show finite variance and no perfect multicollinearity. Travel distance and travel direction are moderately correlated, so coefficient magnitudes and significance should be interpreted cautiously.

A Note on Linear Relationships - Linear regression assumes linear relationships between input and output variables. Due to the size of our data set, the substantial period of time it covers, and the variability around the slight linear trend, non-linear patterns may be masked.

Model Results

Interpretation - The final model (Model 4) is statistically significant overall (F-test $p < 2.2e\text{-}16$), though it explains only a small proportion of the variance in point differentials (Adjusted R-squared = 0.005). The coefficient for distance traveled is-0.0002 ($p > 0.05$), indicating no statistically significant effect on point differential. Practically, even the longest trip of approximately 3,000 miles would be associated with a decrease in point differential of just 0.6 points, which is negligible in a typical game context. In contrast, traveling West to East shows a positive and statistically significant coefficient of 0.959 ($p < 0.01$), suggesting that away teams traveling in this direction gain nearly a 1-point advantage compared to those in the baseline category: “Neither”, potentially due to better alignment with circadian rhythms or time zone effects. East to West travel does not have a statistically significant effect in the full model. Game attendance has a small but statistically significant negative effect on point differential (-0.0002, $p < 0.01$), implying that larger crowds may slightly disadvantage the away team, for example, an increase of 10,000 spectators is associated with a decrease of about 2 points in away team point differential. Lastly, the positive coefficient on year (0.069, $p < 0.01$) indicates an upward trend in point differentials over time, perhaps reflecting changing competitive dynamics or scoring trends.

Overall, the results suggest that while geographic orientation and temporal trends influence competitive outcomes, travel distance alone is not a primary driver of game performance.

Scope & Limitations - This analysis is descriptive in nature and does not support causal claims about the effect of travel on NBA game outcomes. Games are not fully independent, as team performance may be correlated across time due to persistent team quality, injuries, and broader league dynamics. Additionally, while geodesic distance captures the physical separation between teams, it does not directly measure fatigue, rest, travel conditions, or circadian disruption, such as back-to-back games or sleep loss from changing time zones. Finally, structural changes in the NBA over the seven-decade sample period, including standardization of courts, more movement of players between teams, and traveling being less strenuous, may influence both the distribution of point differentials and their relationship with travel-related variables.

Conclusion

Our analysis finds little evidence that travel distance alone is meaningfully associated with point differential in NBA regular season games. While the estimated effect of distance is negative, it is small in magnitude and not statistically distinguishable from zero. In contrast, travel direction exhibits more consistent associations, with West-to-East travel linked to higher away-team point differentials, suggesting that geographic position and time-zone alignment may play a larger role than distance itself.

Citations

Effect of Travel and Rest on Performance of Professional Basketball Players. (1997). *Sleep*. https://doi.org/10.1093/sleep/20.5.366

Smith, R., Guilleminault, C., & From, B. (1996). Peak athletic performance time and circadian advantage in professional athletes. *Sleep Research, 25*, 53.

Appendix

Link to the Dataset : https://www.kaggle.com/datasets/eoinamoore/historical-nba-data-and-player-box-scores

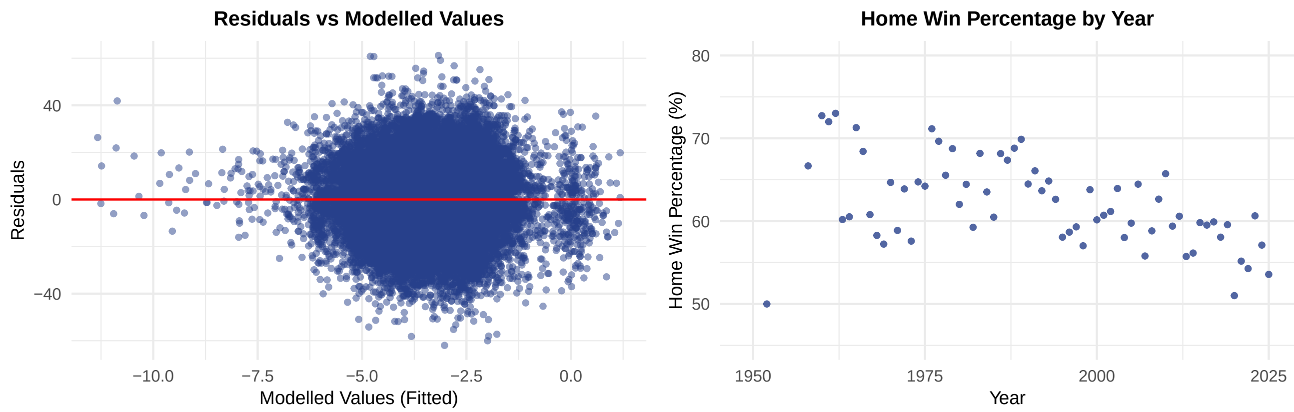

Figure 2: Residuals and home win percentage over time.

Table 2: Covariance Matrix of Model Variables

Variable	dist_miles	year	attendance	travel_direction	point_diff
dist_miles	460527.15	-585.82	-121372.94	315.70	-29.13
year	-585.82	282.15	49969.77	-1.29	10.72
attendance	-121372.94	49969.77	25760567.05	-243.67	-947.93
travel_direction	315.70	-1.29	-243.67	0.41	-0.07
point_diff	-29.13	10.72	-947.93	-0.07	175.84

Table 3: Correlation Matrix of Model Variables

Variable	dist_miles	year	attendance	travel_direction	point_diff
dist_miles	1.00	-0.05	-0.04	0.72	0.00
year	-0.05	1.00	0.59	-0.12	0.05
attendance	-0.04	0.59	1.00	-0.07	-0.01
travel_direction	0.72	-0.12	-0.07	1.00	-0.01
point_diff	0.00	0.05	-0.01	-0.01	1.00